TEAM 1 — SMART CLASSROOM ENERGY MANAGEMENT

Dataset-selection proposal for faculty approval (deadline 9 Oct 2026) · B.Tech CSE, Semester 7 · Prepared 8 Oct 2026

Tags used: **[V]** verified from the dataset authors' own repository/paper, or reproduced by us from the raw files · **[I]** inference · **[D]** derived variable · **[S]** simulated / proposed.

1. Project Objective

Energy is wasted when classroom lighting or air-conditioning runs while the room is empty. We will build an IoT pipeline that monitors occupancy, temperature, ambient light and electrical energy, estimates occupancy and energy demand with ML, and uses the result to switch classroom lighting and AC through MQTT.

Pipeline (as required): **Real-world dataset replayed by a Python IoT simulator → MQTT broker → Node-RED → SQLite → Node-RED dashboard → ML analytics → decision engine → MQTT broker → simulated Light / AC actuators.**

2. Required Data

Required by the problem statement	Available in the selected dataset (ROBOD, lecture rooms)
Occupancy	Yes [V]: camera-based `occupant_presence` (0/1) and `occupant_count`, plus `wifi_connected_devices`
Temperature	Yes [V]: `air_temperature` (°C)
Ambient light	Yes [V]: `illuminance` (lux), with a caveat (see Section 6)
Electrical power / energy	Yes [V]: `lighting_energy`, `plug_load_energy`, `fcu_fan_energy` (kWh); `chilled_water_energy` is thermal cooling energy (kWh)
Timestamp	Yes [V]: `YYYY-MM-DD HH:MM +08:00`, 5-minute interval

3. Candidate Dataset Analysis

Nine candidates were examined. Class: **A** = all four variables real, **B** = one missing but legitimately derivable/simulable, **C** = several missing, **D** = does not support the project.

#	Dataset (original source)	Size · interval · period	Occ.	Temp	Light	Power	Class
1	**ROBOD** — NUS Singapore, Building Simulation 2022	5 rooms (2 lecture rooms); 5 min; 2021-09-07 to 2021-12-23; 8,352 rows per lecture room [V]	Yes (camera count + presence)	Yes	Yes (lux)	Yes (lighting, plug, fan kWh; chilled-water kWh)	**A**
2	UCI Occupancy Detection	20,560 instances; ~1 min; period UNVERIFIED; 1 office room	Yes (0/1)	Yes	Yes	No	B

#	Dataset (original source)	Size · interval · period	Occ.	Temp	Light	Power	Class
3	Aalborg 6 office rooms — Zenodo, Data in Brief 2024	6 rooms; 5 min; 2023-02-27 to 2023-12-31 (~88.7k timestamps [I])	Yes, but 34–54% missing	Yes	Yes + light on/off	No room-level power (only shared AHU)	B
4	UCI Room Occupancy Estimation	≈10.1k rows (exact count UNVERIFIED); 30 s sensors; 4 days; 1 room	Yes (count 0–3)	Yes	Yes	No (no HVAC in use)	B
5	CU-BEMS — Chulalongkorn Univ., Scientific Data 2020	7-storey building, 33 zones; 1 min; 2018-07-01 to 2019-12-31	**No**	Yes	Yes	Yes (AC, lighting, plug in kW)	B
6	UCI Appliances Energy Prediction	19,735 rows; 10 min; ~4.5 months; a house	No	Yes	No lux (lights in Wh)	Yes (Wh)	C
7	SDU Odense occupant counts — Scientific Data 2019	44 days (Mar 2018–Apr 2019); columns UNVERIFIED	Yes	Yes (IEQ)	UNVERIFIED	No energy mentioned	C
8	Building Data Genome 2 / ASHRAE GEPIII	3,053 meters, 1,636 buildings; hourly; 2016–2017	No	weather only	No	Yes (whole-building)	D
9	Murcia smart campus — IEEE DataPort	4 buildings; hourly; CO2 + weather	Some (people counters)	Temp min/max (weather)	No	No	C

Original sources:

1. ROBOD: https://github.com/ideas-lab-nus/robod · DOI https://doi.org/10.6084/m9.figshare.19234530.v7 · paper https://doi.org/10.1007/s12273-022-0925-9
2. https://archive.ics.uci.edu/dataset/357/occupancy+detection (CC BY 4.0)
3. https://zenodo.org/records/10673763 (CC BY 4.0) · paper https://doi.org/10.1016/j.dib.2024.110326

4. https://archive.ics.uci.edu/dataset/864/room+occupancy+estimation
5. https://doi.org/10.1038/s41597-020-00582-3 · data site https://sgrudata.github.io/
6. https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction (CC BY 4.0)
7. https://doi.org/10.1038/s41597-019-0274-4
8. https://github.com/buds-lab/building-data-genome-project-2 · arXiv 2006.02273
9. https://ieee-dataport.org/documents/weakly-supervised-occupancy-prediction-hvac-optimization-digital-twins-smart-campuses

Key reasons for non-selection: candidates 2 and 4 are single rooms with no energy data; 3 has no room-level power and large occupancy gaps, and is a heavy 87.6 MB / 200+ column file; 5 has no occupancy; 6–9 miss multiple required variables or are not classroom/room-level.

4. Dataset Comparison

Weights: sensor coverage 30%, ML potential 15%, data quality 15%, IoT-simulation suitability 15%, academic credibility 10%, implementation simplicity 10%, accessibility/licence 5%. Scores (1–10) are our own judgements from the verified facts; the arithmetic was computed by script.

Dataset	Coverage	ML	Quality	IoT sim	Credibility	Simplicity	Access	**Weighted**
ROBOD	9	8	8	9	8	9	7	**8.50**
UCI Occupancy Detection	6	6	7	8	9	10	10	7.35
Aalborg 6 offices	7	8	6	6	9	4	10	6.90
UCI Room Occupancy Estimation	6	6	6	7	8	9	10	6.85
CU-BEMS	6	7	6	7	9	5	7	6.55
UCI Appliances Energy	3	6	8	7	9	9	10	6.35
SDU Odense	4	6	6	6	9	6	7	5.75
BDG2 / ASHRAE	2	7	7	4	10	4	8	5.10
Murcia smart campus	3	5	5	4	6	6	5	4.45

Coverage rule used for every candidate: all four variables real = 9–10; three real plus a partial proxy for the fourth = 7; three real, no proxy = 6; two or three = 4–5; one or none = 2–3. ROBOD loses points for caveats (energy is per-interval kWh, the AC cooling energy is thermal, and the light sensor behaves unusually in lecture rooms). The simple UCI datasets score well on simplicity but cannot show any energy saving without inventing the entire power signal. ROBOD leads by more than a full point, so the choice does not depend on small scoring differences.

5. Recommended Dataset

ROBOD — Room-level Occupancy and Building Operation Dataset (Tekler et al., National University of Singapore; *Building Simulation* 15(12), 2022). We will use **Room 1 (lecture room)** as the main classroom and **Room 2 (second lecture room)** for a second room and for generalisation tests.

Why it fits: it is the only candidate that contains occupancy, temperature, ambient light and metered lighting/fan/plug energy **in the same rooms, at the same timestamps**, and the rooms are real university lecture rooms.

Verified facts [V] (downloaded from the authors' GitHub repository `ideas-lab-nus/robod`, files `combined_Room1.csv` and `combined_Room2.csv`, and checked against the authors' sensor table):

- 8,352 rows × 28 columns per lecture room = 29 days × 288 five-minute steps. Together with Rooms 3–5 the paper reports 181 room-days.
- Coverage: 7 Sep–1 Oct 2021 and 9–23 Dec 2021 (non-contiguous), weekdays only.
- Missing values are tiny: Room 1 — illuminance 0.17%, chilled water/FCU fan 0.12%; Room 2 — illuminance 0.36%; everything else 0%.
- Occupied (presence = 1) in 21.1% of Room 1 rows and 28.4% of Room 2 rows; peak count 38 (Room 1) and 22 (Room 2).
- Units from the authors' table: temperature °C, illuminance lux, Wi-Fi devices count, energy kWh.
- Licence: not stated on the GitHub page and the Figshare page blocked automated access, so the licence is **UNVERIFIED**. The authors ask users to cite the paper; we will do so.

What must be derived or simulated: actuator states (light on/off, AC mode) and all control commands are **not** in the dataset; the controlled-energy scenario is a modelled counterfactual.

6. Dataset Variable Mapping

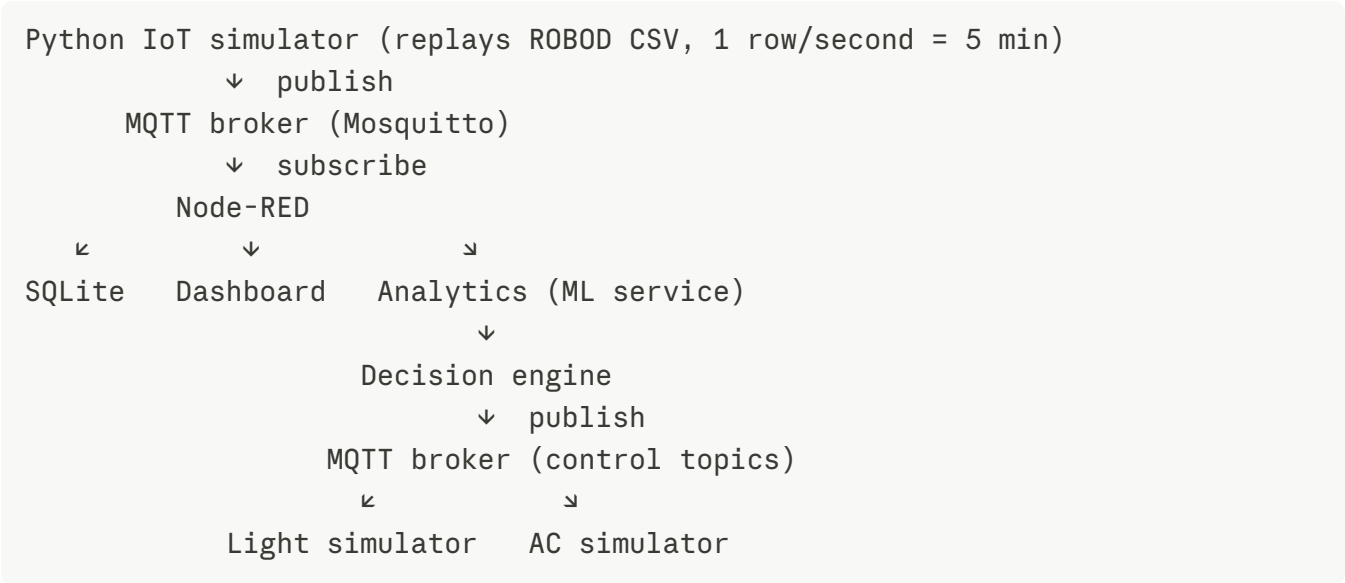
Actual columns in the CSV (Room 1/2): `timestamp, voc, sound_pressure_level, indoor_relative_humidity, air_temperature, illuminance, pm2.5, indoor_co2, wifi_connected_devices, ceiling_fan_energy, lighting_energy, plug_load_energy, chilled_water_energy, fcu_fan_energy, temp_setpoint, fcu_fan_speed, supply_air_pressure, supply_air_temperature, baromatic_pressure, dry_bulb_temp, global_horizontal_solar_radiation, wind_direction, wind_speed, outdoor_co2, rainfall_raw, outdoor_relative_humidity, occupant_presence, occupant_count`.

IoT field	Source column / rule	Unit	Class
timestamp	`timestamp`	ISO 8601, +08:00	Real
occupant_presence	`occupant_presence` (camera)	0/1	Real
occupant_count	`occupant_count` (camera)	persons	Real
wifi_devices	`wifi_connected_devices`	count	Real
temperature_c	`air_temperature`	°C	Real
illuminance_lux	`illuminance`	lux	Real
lighting_kwh	`lighting_energy`	kWh per 5 min	Real
fcu_fan_kwh	`fcu_fan_energy`	kWh per 5 min	Real
chilled_water_kwh	`chilled_water_energy`	kWh (thermal)	Real
plug_kwh	`plug_load_energy`	kWh per 5 min	Real
setpoint_c	`temp_setpoint`	°C	Real

IoT field	Source column / rule	Unit	Class
lighting_kw (avg)	lighting_kwh × 12	kW	Derived [D][I] (assumes per-interval energy; values are small and non-cumulative)
light_on_baseline, ac_active_baseline	threshold on lighting_kwh; chilled_water or fan > 0	binary	Derived [D]
hour, day_of_week, occupancy profile	from timestamp	—	Derived [D]
predicted_occupancy, predicted_energy	ML output	—	Derived (ML)
light_status, ac_status, control commands, alerts	decision engine + actuator simulator	ON/OFF/ECO	**Simulated [S]**
baseline vs controlled energy	replay model (Section 14)	kWh	**Simulated [S]**

Two caveats we will state openly: (1) AC cooling is chilled-water **thermal** energy while fan energy is **electrical**; we keep them as separate series and never add them as one number. (2) In the lecture rooms the light sensor reads low (Room 1 maximum 92 lux; median 0 lux when occupied, probably projector/dimmed-lighting use) and is weakly related to lighting energy. Room 1 therefore cannot demonstrate daylight dimming; Room 2 (maximum ~2,015 lux) can.

7. Proposed IoT Architecture



Component	Responsibility
Python simulator	Reads ROBOD rows in time order and publishes each as a JSON sensor reading; accelerated replay
Mosquitto	Message routing between simulator, Node-RED and actuators
Node-RED	Subscribe, validate, store, call ML, run decision rules, publish commands, serve dashboard
SQLite	Stores telemetry, predictions, decisions
ML service (small Flask API, model trained offline in scikit-learn)	Returns occupancy estimate and predicted energy for a reading
Actuator simulator (Python)	Subscribes to control topics, changes simulated state, publishes acknowledgement

8. MQTT Communication Design

Topic	Publisher → Subscriber	QoS	Retained
`smartclass/room1/telemetry`	Simulator → Node-RED	1	No
`smartclass/room1/status` (online/offline, Last Will)	Simulator → Node-RED	1	Yes
`smartclass/room1/analytics`	Node-RED → Dashboard/DB	0	No
`smartclass/room1/control/light`	Node-RED → Light simulator	1	Yes
`smartclass/room1/control/ac`	Node-RED → AC simulator	1	Yes
`smartclass/room1/actuator/light` and `/actuator/ac` (state ack)	Simulators → Node-RED	1	Yes
`smartclass/room1/alert`	Node-RED → Dashboard	1	No

Room 2 uses the same structure with `room2`. Publishing frequency: one telemetry message per replayed step (default 1 s = 5 simulated minutes, configurable). One JSON message per step keeps all fields time-aligned and is easier to explain than many separate topics.

Example telemetry payload (values taken from the real Room 1 row at 2021-09-07 09:10, rounded):

```
{"room":"room1","ts":"2021-09-
07T09:10:00+08:00","occupant_presence":1,"occupant_count":23,
 "wifi_devices":20,"temperature_c":26.52,"illuminance_lux":18.65,
 "lighting_kwh":0.20507,"fcu_fan_kwh":0.031,"chilled_water_kwh":1.0}
```

Example control payload: `{"room":"room1","ts":"...","cmd":"OFF","reason":"unoccupied_15min"}`.

9. Database Design

SQLite (the problem statement allows SQLite or MySQL; SQLite needs no server and is easy to demo and debug), accessed through `node-red-node-sqlite`.

Table `telemetry`: `id INTEGER PRIMARY KEY AUTOINCREMENT`, `room_id TEXT`, `ts TEXT` (ISO 8601 with +08:00), `occupant_presence INTEGER`, `occupant_count INTEGER`, `wifi_devices INTEGER`, `temperature_c REAL`, `illuminance_lux REAL`, `lighting_kwh REAL`, `fcu_fan_kwh REAL`, `chilled_water_kwh REAL`, `UNIQUE(room_id, ts)`.

Table `decisions`: `id`, `room_id`, `ts`, `pred_occupancy`, `pred_energy_kwh`, `light_cmd`, `ac_cmd`, `reason`, `baseline_kwh`, `controlled_kwh`.

Example record: `(room1, 2021-09-07T09:10:00+08:00, 1, 23, 20, 26.52, 18.65, 0.20507, 0.031, 1.0)`. Timestamps are stored as ISO text, which sorts correctly and preserves the time-zone offset.

10. Node-RED Integration

`mqtt in` (telemetry) → `json` → `function` (validate: required fields present, values inside the sensor ranges in the authors' table, drop duplicates) → `sqlite` (insert) → `http request` (ML service) → `function` (decision engine rules) → `switch` / `rbe` (send only on state change) → `mqtt out` (control topics) → actuator simulators → `mqtt in` (state acknowledgement) → dashboard. Supporting nodes: `ui_gauge`, `ui_chart`, `ui_text`, `ui_led`, `ui_toast`/notification, `inject` for tests, `catch` and `debug` for errors. Everything is a visible flow, so each member can explain and modify a stage live.

11. Dashboard Design

- **Current state:** occupancy (count + present/absent), temperature, illuminance, lighting kWh and fan/cooling kWh, Light status, AC status.
- **Analytics charts (only the useful ones):** (1) energy vs time with occupancy count overlaid; (2) hourly occupancy profile; (3) predicted vs actual energy; (4) cumulative baseline vs controlled energy.
- **Automation panel:** current decision and its reason, energy-saving status, last 10 decisions, alert/anomaly list, manual override switch.

12. Analytics / ML Plan

Principle: simple, explainable models first; no deep learning.

Task	Input features	Target	Algorithm	Metric
M1 Occupancy estimation (works without the camera)	wifi_devices, temperature, illuminance, hour, day_of_week	`occupant_presence`	Logistic regression / decision tree baseline → Random Forest	F1, precision/recall, confusion matrix; compare against majority-class baseline
M2 Energy demand	occupant_count, hour, day_of_week, temperature, illuminance, setpoint	next-step `lighting_kwh` and `chilled_water_kwh`	Random Forest / Gradient Boosting regressor	MAE, RMSE vs mean baseline
M3 Imminent occupancy	hour × weekday occupancy-probability table learned from training days	P(occupied in next 30 min)	Group-by profile (no model needed)	Agreement with ground truth on held-out days
M4 Anomaly (advanced, optional)	energy, occupancy, hour	normal / abnormal	Isolation Forest or z-score by hour	Manual review of flagged cases

Evaluation always **holds out whole days** (GroupKFold by date) because neighbouring 5-minute rows are almost identical.

Preliminary feasibility check (ours, run during dataset verification; not final project results): with whole days held out and a Random Forest, estimating presence from Wi-Fi + temperature + light + time gave accuracy 0.88 / F1 0.69 (Room 1) and 0.88 / 0.77 (Room 2), against majority-class accuracy 0.79 and 0.72. Without Wi-Fi, F1 dropped to 0.43 and 0.60, so the Wi-Fi device count is the key proxy sensor. Predicting lighting energy from occupancy and time cut MAE from 0.055 to 0.040 kWh (Room 1) and from 0.024 to 0.016 kWh (Room 2) versus a mean baseline. **Forecasting future presence (30 and 60 min**

ahead) did not beat the simple persistence baseline with only 29 days of data. The design therefore uses ML where it helps (current-occupancy estimation, energy demand) and a transparent profile-plus-rules approach for pre-conditioning, and we will report the persistence comparison honestly.

13. Decision & Automation Plan

Inputs: occupancy (sensor or ML estimate), P(next 30 min), temperature, illuminance, hour. Rules use hysteresis and minimum on/off times to avoid chattering.

Condition	Light	AC
Unoccupied ⩾ 15 min and P(next 30 min) < 0.3	OFF	ECO (setback), full OFF after 30 min
Occupied	ON (or dim if enough ambient light)	Comfort setpoint
Unoccupied but P(next 30 min) ⩾ 0.5 (imminent occupancy)	stay OFF	Pre-cool to comfort setpoint
Temperature > setpoint + 1.5 °C while occupied	—	Boost
Temperature below the lowest dataset setpoint (22 °C)	—	ECO (avoid over-cooling)
Ambient light ⩾ calibrated threshold (default 300 lux, to be calibrated on Room 2)	Dim/OFF	—
Lighting or cooling energy high while room empty (anomaly)	Force OFF	Alert + ECO
Telemetry missing > 10 s	Hold last state; during 08:00–20:00 default ON	Hold; raise alert

A manual-override switch always wins. These rules are the safe baseline; ML outputs feed the occupancy estimate and the pre-conditioning probability, so analytics always lead to a concrete MQTT command.

14. Expected Energy-Saving Demonstration

Method only; no savings are claimed before experiments are run.

- **Baseline:** replay a day with the energy exactly as recorded (no control).
- **Controlled scenario:** replay the same day through the decision engine. For each step where the light is commanded OFF, controlled lighting energy = 0 (or a declared standby value); where AC is in ECO, controlled cooling energy = *f* × recorded, with *f* declared in advance and varied (for example 0.5, 0.75) as a sensitivity test.
- **Energy saved** = Σ(baseline − controlled); **% saving** = saved ÷ baseline, reported per load (lighting, fan, cooling) and never mixing thermal with electrical energy.
- **Comfort cost:** count occupied steps where the light was OFF or the temperature exceeded setpoint + 2 °C (false-OFF rate).
- **Sanity ceiling (computed by us from the raw files):** the share of recorded energy during steps with no occupant detected — Room 1: lighting 36.1%, cooling 54.2%, fan 55.6%; Room 2: lighting 35.8%, cooling 37.3%, fan 42.5%. This is an upper bound on opportunity, not an achievable saving (it includes start-up and pre-cooling).

15. Backup Dataset

UCI Occupancy Detection (weighted score 7.35): https://archive.ics.uci.edu/dataset/357/occupancy+detection · CC BY 4.0 · 20,560 instances · columns `date, Temperature, Humidity, Light, CO2, HumidityRatio, Occupancy` · no missing values. It is simple, credible and instantly accessible, but it has **no power data**, so lighting/AC power would be fully simulated (rated power × simulated state) and labelled as such. It would be used only if ROBOD becomes unusable, for example because of a licensing problem. CU-BEMS (real power, no occupancy) and the Aalborg dataset are reserve options.

16. Conclusion and Faculty Approval Request

ROBOD is the only examined dataset that provides real lecture-room occupancy, temperature, ambient light and metered lighting/fan energy at 5-minute resolution, from a peer-reviewed source, in small clean CSV files that map directly onto the MQTT → Node-RED → SQLite → Dashboard → ML → Decision → Actuator architecture. Its limits (29 days, weekdays only, no actuator states, unusual light-sensor behaviour, licence not confirmed) are documented and handled in the design.

We request approval to use ROBOD (Room 1 primary, Room 2 secondary) as the Team 1 dataset, with UCI Occupancy Detection as backup.

Team Prep Pack (for our team — not part of the faculty submission)

A. 60-second viva explanation

Classrooms waste energy when the lights and AC stay on in an empty room. We picked the ROBOD dataset from NUS because it has real lecture-room data every 5 minutes: occupancy from cameras, temperature, light level, and metered lighting and fan energy. A Python script replays the CSV like a live sensor and publishes each reading as JSON over MQTT to a Mosquitto broker. Node-RED subscribes, validates the data, stores it in SQLite and shows it on the dashboard. It also calls a small ML service: a Random Forest estimates occupancy from the Wi-Fi device count, temperature, light and time of day, and a regressor predicts energy demand. A rule-based decision engine then uses that output. If the room has been empty for 15 minutes and is not expected to fill soon, Node-RED publishes OFF for the light and ECO for the AC; if the usual timetable pattern says people are about to arrive, it pre-cools. Simulated light and AC subscribers act on those topics and send their state back. To show savings, we replay the same day twice, once as recorded and once with our control, and compare energy while counting comfort misses. We will report numbers only after running it.

B. Faculty questions and answers

1. **Why did you choose this dataset?** It is the only one we found with occupancy, temperature, light and metered energy in the same real lecture rooms at the same timestamps. It is peer-reviewed (NUS, *Building Simulation* 2022) and the files are small and clean.
2. **Does it contain all required sensor parameters?** Yes, all four are real. Two caveats: AC cooling is chilled-water thermal energy plus electrical fan energy, not one AC power meter; and the Room 1 light sensor reads low (maximum 92 lux).
3. **Which variables are real and which are simulated?** Real: occupancy (camera), Wi-Fi count, temperature, illuminance, lighting/fan/plug/chilled-water energy, setpoint. Derived: kW, hour/day features, predictions. Simulated: light and AC states, control commands, and the controlled-energy scenario.

4. **Where exactly is the IoT component?** The Python publisher is the simulated IoT sensor node (with a Last Will message), and the Light and AC simulators are IoT actuator nodes. Both talk to the broker over MQTT. The faculty document allows a real-world dataset as the sensor source.
5. **Why MQTT?** It is a lightweight publish/subscribe protocol that decouples sensors, Node-RED and actuators, and gives QoS, retained messages and Last Will.
6. **What does Node-RED do?** It integrates and automates: subscribes to telemetry, validates, writes to SQLite, calls the ML service, runs the decision rules, publishes commands and serves the dashboard.
7. **Why this ML algorithm?** Random Forest handles non-linear, mixed features with little tuning and gives feature importance. We start with logistic regression/decision tree baselines so the choice is justified, not assumed.
8. **What is your target variable?** `occupant_presence` for occupancy estimation, and next-step `lighting_energy` / `chilled_water_energy` for demand.
9. **How will you evaluate the model?** F1, precision/recall and confusion matrix for occupancy; MAE/RMSE for energy; always holding out whole days and comparing with majority-class, mean and persistence baselines.
10. **How will the model control the classroom?** The ML output feeds the Node-RED decision rules, which publish retained commands to `control/light` and `control/ac`; the simulators change state and acknowledge.
11. **How will you demonstrate energy savings?** Replay the same day as baseline (as recorded) and as controlled (our decisions), with declared assumptions for how much energy each command removes, then report saved kWh, % and the comfort-miss count. No numbers are claimed yet.
12. **What happens when occupancy data is unavailable?** The M1 model estimates it from Wi-Fi count, temperature, light and time. In our preliminary check this gave F1 0.69 / 0.77 with Wi-Fi but only 0.43 / 0.60 without it, so Wi-Fi is key; if telemetry stops, the system holds its last state and raises an alert.
13. **Is the system real-time?** It is near-real-time and time-compressed: one 5-minute row is published per second. The architecture would work on live data, but this is a simulation, not a deployed system.
14. **Why didn't you choose another dataset?** UCI Occupancy Detection and Room Occupancy Estimation have no energy data; Aalborg has no room-level power and large occupancy gaps; CU-BEMS has no occupancy; UCI Appliances is a house; BDG2 is whole-building energy only.

15. **What happens after approval?** Install Mosquitto, Node-RED and SQLite; write the simulator; train M1/M2 offline; build the flows and dashboard; implement the decision rules; run the baseline-vs-controlled experiments; rehearse the viva.

C. Checklist for tomorrow

MUST HAVE

- This proposal open (or printed), especially Sections 3–6.
- The links open: https://github.com/ideas-lab-nus/robod and the paper DOI https://doi.org/10.1007/s12273-022-0925-9.
- Every member can state which variables are real, derived and simulated.

SHOULD HAVE

- A screenshot of `combined_Room1.csv` loaded in pandas (8,352 rows × 28 columns).
- The 60-second explanation rehearsed by at least two members.
- Ask faculty to confirm that using ROBOD for academic work with citation is acceptable (licence UNVERIFIED).

OPTIONAL

- Mosquitto and Node-RED installed; a ten-line paho script publishing one real row.
- The pandas snippet behind the "energy while unoccupied" figures.

D. Verification log

Verified from original sources or reproduced from raw data: ROBOD GitHub README (data format, files, period, 5-min interval, +08:00 timestamps); the authors' sensor/unit table; actual CSV columns, row counts, missing values and sample rows for Rooms 1 and 2; Aalborg Zenodo record (CC BY 4.0) and its Data in Brief tables; UCI pages for Occupancy Detection (CC BY 4.0) and Appliances Energy (CC BY 4.0); CU-BEMS and BDG2 abstracts.

UNVERIFIED (labelled as such above): ROBOD licence (Figshare page blocked automated access); UCI Occupancy Detection recording period; exact row count and full column list of UCI Room Occupancy Estimation (taken from secondary listings); CU-BEMS licence and file layout; SDU Odense columns/size/licence; BDG2 licence; Murcia access conditions; the Aalborg timestamp count (derived from the paper's tables).

Our own analysis (reproducible with pandas/scikit-learn, not published results): energy-while-unoccupied shares and the preliminary ML feasibility check. They are not project results.

FINAL RECOMMENDATION

STATUS: GO

Primary Dataset: ROBOD — Room-level Occupancy and Building Operation Dataset (NUS), Room 1 primary, Room 2 secondary

Backup Dataset: UCI Occupancy Detection (power would have to be simulated)

Overall Score: 8.50 / 10 (next best 7.35)

Dataset Approval Confidence: 8 / 10

Top 3 Strengths:

1. All four required variables are real and time-aligned in actual lecture rooms (camera occupancy, temperature, lux, metered energy).
2. Peer-reviewed source with clean 5-minute CSVs (8,352 rows per lecture room, under 0.4% missing values) that replay naturally as MQTT messages.
3. Real energy consumed while rooms were empty gives a credible, measurable baseline for the saving demonstration.

Top 3 Risks:

1. Only 29 weekdays per lecture room; no weekends, and the days are non-contiguous, so forecasting is limited (it did not beat persistence in our check).
2. The Room 1 light sensor reads low and is weakly related to lighting energy, so daylight dimming needs Room 2 or a calibrated threshold.
3. Licence not confirmed, and AC energy is thermal plus fan electrical rather than a single AC power meter.

Critical Missing Variables: Actuator states (light on/off, AC mode) and a direct AC electrical power reading are not in the dataset; they are derived or simulated and will be labelled that way.

What We Need Faculty to Approve Tomorrow: ROBOD (Room 1 primary, Room 2 secondary) as the Team 1 dataset, with simulated actuators and a modelled baseline-vs-controlled energy comparison, and UCI Occupancy Detection as the backup.